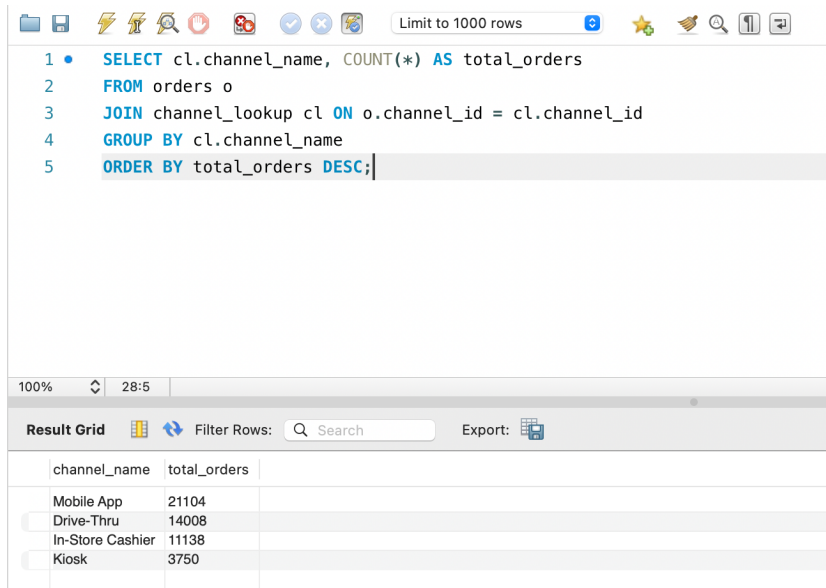


3 Basic Queries

Problem Statement: Which ordering channel do Starbucks customers use the most and how does channel preference vary across the four available options?



```
1 • SELECT cl.channel_name, COUNT(*) AS total_orders
2 FROM orders o
3 JOIN channel_lookup cl ON o.channel_id = cl.channel_id
4 GROUP BY cl.channel_name
5 ORDER BY total_orders DESC;
```

channel_name	total_orders
Mobile App	21104
Drive-Thru	14008
In-Store Cashier	11138
Kiosk	3750

Key finding: Mobile App dominates with 21,104 orders (42%) confirming Starbucks' digital strategy is working, while Kiosk remains significantly underutilized at just 3,750 orders.

To further explain :

1. I wrote a basic SELECT query joining orders and channel_lookup to count how many orders were placed through each channel this confirms the dataset is loaded correctly and the FK relationship between the two tables works.
2. I used GROUP BY on channel_name to aggregate order counts per channel, and ORDER BY DESC to rank channels from most to least used.
3. Mobile App is the most popular channel with 21,104 orders (42% of all orders) showing Starbucks customers strongly prefer digital ordering over in-person methods.
4. Drive-Thru comes second with 14,008 orders, while In-Store Cashier has 11,138 together these three channels account for over 92% of all orders.
5. Kiosk is the least used channel with only 3,750 orders suggesting customers either don't find kiosks convenient or are unaware of them.

Problem Statement: What is the gender distribution of Starbucks customers and does the brand appeal equally across all gender groups?

```
1 • SELECT customer_gender, COUNT(*) AS total_customers
2 FROM customer
3 GROUP BY customer_gender
4 ORDER BY total_customers DESC;
```

100%	31:4
Result Grid   Filter Rows: Search Export: 	
customer_gender	total_customers
Female	6808
Male	6700
Non-binary	754
Prefer not to say	726

Key finding: Starbucks' customer base is nearly gender-balanced, Female (6,808) and Male (6,700) are almost equal, with 10% identifying as Non-binary or preferring not to disclose.

To further explain :

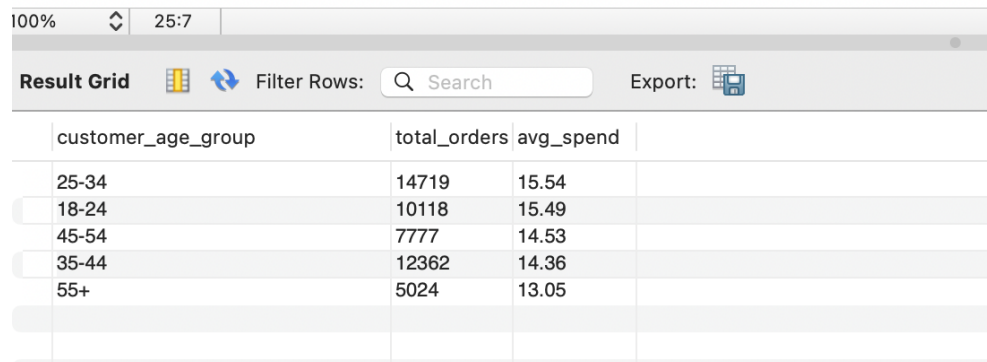
1. I wrote a simple SELECT query on the customer table to count how many unique customers belong to each gender category, this confirms customer demographic data loaded correctly with no nulls or missing groups.
2. I used GROUP BY on customer_gender to aggregate customer counts per gender, and ORDER BY DESC to rank from largest to smallest group.
3. Female customers are the largest group at 6,808 followed closely by Male at 6,700 and the difference is only 108 customers, meaning the customer base is almost perfectly balanced between the two.

4. Non-binary customers account for 754 and "Prefer not to say" for 726 together about 10% of the total customer base, showing Starbucks' dataset captures gender inclusively beyond binary categories.

5. The near-equal split between Female and Male suggests Starbucks has broad demographic appeal and is not skewed toward any single gender group.

Business Question: Which customer age group spends the most per order and where should Starbucks focus its marketing efforts?

```
1 • SELECT customer_age_group,  
2         COUNT(*) AS total_orders,  
3         ROUND(AVG(o.total_spend), 2) AS avg_spend  
4 FROM orders o  
5 JOIN customer c ON o.customer_id = c.customer_id  
6 GROUP BY customer_age_group  
7 ORDER BY avg_spend DESC;
```



100% 25:7

Result Grid Filter Rows: Search Export:

customer_age_group	total_orders	avg_spend
25-34	14719	15.54
18-24	10118	15.49
45-54	7777	14.53
35-44	12362	14.36
55+	5024	13.05

Key Finding: The 25-34 age group leads in both order volume (14,719) and average spend (\$15.54), while 55+ has the lowest spend at \$13.05 suggesting younger customers drive the most value for Starbucks.

To further explain:

- I wrote a SELECT query joining the orders and customer tables to calculate average spend and total orders per age group, this confirms the JOIN between both tables works correctly and demographic data is usable.

- I used GROUP BY customer_age_group to aggregate spend per age group and ORDER BY avg_spend DESC to rank from highest to lowest spender so the most valuable age segment appears first.
- The 25-34 group spends the most at \$15.54 per order on average and also places the most orders at 14,719 making them Starbucks' single most valuable segment on both dimensions simultaneously.
- The 18-24 group comes second in spend at \$15.49 nearly identical to 25-34 but places fewer orders at 10,118, suggesting younger customers spend similarly per visit but visit less frequently.
- The 55+ group has the lowest average spend at \$13.05 and fewest orders at 5,024 suggesting simpler, less customized orders. Starbucks should focus premium upselling on the 25-34 and 18-24 segments and loyalty rewards on 35-44 who have high volume at 12,362 orders despite moderate spend.

2 Avance Query

Problem Statement: Identify which Starbucks ordering channels have above-average wait times and below-average satisfaction scores, to pinpoint where operational improvements are most needed.

```

1 • CREATE OR REPLACE VIEW underperforming_orders AS
2   SELECT
3     o.order_id,
4     c.customer_id,
5     o.total_spend,
6     o. fulfillment_time_min,
7     o.customer_satisfaction,
8     cl.channel_name
9   FROM orders o
10  JOIN customer c ON o.customer_id = c.customer_id
11  JOIN channel_lookup cl ON o.channel_id = cl.channel_id;
12
13 • SELECT
14   channel_name,
15   COUNT(*) AS underperforming_count,
16   ROUND(AVG(total_spend), 2) AS avg_spend,
17   ROUND(AVG(fulfillment_time_min), 2) AS avg_fulfillment_time,
18   ROUND(AVG(customer_satisfaction), 2) AS avg_satisfaction
19  FROM underperforming_orders
20  WHERE fulfillment_time_min > (
21    SELECT AVG(fulfillment_time_min) FROM orders
22  )
23  AND customer_satisfaction < (
24    SELECT AVG(customer_satisfaction) FROM orders
25  )
26  GROUP BY channel_name
27  ORDER BY underperforming_count DESC;

```

75%

37:27

Result Grid

Filter Rows:

Search

Export:

channel_name	underperforming_co...	avg_spend	avg_fulfillment_ti...	avg_satisfacti...	
Drive-Thru	5031	12.34	6.35	2.22	
Mobile App	3059	18.11	5.47	2.33	
Kiosk	482	12.44	5.23	2.52	
In-Store Cashier	343	12.59	4.99	2.36	

Key Finding : Drive-Thru leads with 5,031 underperforming orders and the lowest satisfaction (2.22/5) making it Starbucks' biggest operational risk.

To further explain :

- I created a VIEW by joining 3 tables **orders, customer, and channel_lookup** so the combined data can be reused across multiple queries without rewriting the joins every time.
- Instead of manually setting a fixed number as the cutoff, I used two subqueries that automatically calculate the current average fulfillment time and satisfaction score from the data itself so the query stays accurate even if the dataset changes.

Instead of using “WHERE fulfillment_time_min > 5.5 -- someone manually typed this number”

I have used “WHERE fulfillment_time_min > (SELECT AVG(fulfillment_time_min) FROM orders)”.

This means the threshold calculates itself from the data automatically. If tomorrow the dataset grows from 50,000 to 500,000 rows, the query still works correctly without anyone changing any numbers manually.

- Drive-Thru had the most underperforming orders at 5,031 nearly double Mobile App with the lowest satisfaction (2.22/5) and longest wait time (6.35 min), indicating clear operational bottlenecks in this channel.
- Mobile App underperformers spent the most on average (\$18.11) yet still reported poor satisfaction (2.33/5) suggesting high-spending customers have higher expectations and are more sensitive to delays.
- In-Store Cashier had the fewest underperforming orders (343) and fastest service (4.99 min) making it the most reliable channel despite being the most traditional.

Business Question: Who are Starbucks' most valuable customers those who spend above average and rate their experience above average simultaneously?

```
1  ● ○ WITH avg_metrics AS (  
2      SELECT  
3          ROUND(AVG(total_spend), 2) AS avg_spend,  
4          ROUND(AVG(customer_satisfaction), 2) AS avg_satisfaction  
5      FROM orders  
6  ),  
7  ○ customer_metrics AS (  
8      SELECT  
9          c.customer_id,  
10         ROUND(SUM(o.total_spend), 2) AS total_spend,  
11         ROUND(AVG(o.customer_satisfaction), 2) AS avg_satisfaction  
12     FROM orders o  
13     JOIN customer c ON o.customer_id = c.customer_id  
14     GROUP BY c.customer_id  
15 )  
16 SELECT  
17     cm.customer_id,  
18     cm.total_spend,  
19     cm.avg_satisfaction,  
20     am.avg_spend AS benchmark_spend,  
21     am.avg_satisfaction AS benchmark_satisfaction  
22 FROM customer_metrics cm, avg_metrics am  
23 WHERE cm.total_spend > am.avg_spend  
24 AND cm.avg_satisfaction > am.avg_satisfaction  
25 ORDER BY cm.total_spend DESC  
26 LIMIT 10;
```

customer_id	total_spend	avg_satisfacti...	benchmark_spe...	benchmark_satisfact...
CUST_14910	206.06	4.00	14.83	3.69
CUST_08642	203.55	4.20	14.83	3.69
CUST_03127	197.86	4.08	14.83	3.69
CUST_01700	193.65	3.78	14.83	3.69
CUST_09463	183.93	3.70	14.83	3.69
CUST_11522	175.14	4.00	14.83	3.69
CUST_04473	172.97	4.13	14.83	3.69
CUST_05531	169.62	3.78	14.83	3.69
CUST_04309	168.55	4.11	14.83	3.69
CUST_09889	168.13	3.88	14.83	3.69

Key Finding: CUST_14910 leads with \$206.06 total spend and 4.00 satisfaction nearly 14x the benchmark spend of \$14.83 while maintaining above average satisfaction of 3.69. All top 10 customers score between 3.70 and 4.20 satisfaction showing high spenders are also consistently happy customers.

To further explain:

- I used two chained CTEs the first CTE `avg_metrics` calculates the overall benchmarks across all 50,000 orders average spend of \$14.83 and average satisfaction of 3.69. The second CTE `customer_metrics` aggregates total spend and average satisfaction per individual customer.
- The final `SELECT` filters only customers who beat both benchmarks simultaneously total spend above \$14.83 AND satisfaction above 3.69. Not just one condition both at the same time.
- `CUST_14910` is the top customer at \$206.06 total spend nearly 14x the average order spend while maintaining a 4.00 satisfaction score above the 3.69 benchmark. This is Starbucks' highest value loyal customer.
- All top 10 customers have satisfaction scores between 3.70 and 4.20 none are perfect 5.0 suggesting even the most satisfied high spenders still have room for experience improvement.
- The benchmark columns (\$14.83 and 3.69) are the same across all rows because they are calculated once from the entire dataset this is the power of the first CTE compute once, reuse everywhere. These customers spend on average 10-14x the benchmark identifying them as Starbucks' most loyal and profitable segment for premium targeting.

Note: These two advanced queries tell opposite stories from the same dataset. The `VIEW` query finds where Starbucks is failing channels with worst wait times and lowest satisfaction. The `CTE` query finds where Starbucks is winning customers who spend the most AND are the most satisfied. Together they give Starbucks a complete picture fix the broken channels first, then double down on retaining the best customers."